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Plugging into the Future: An Exploration of Electricity Consumption Patterns

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INTRODUCTION

Plugging into the Future: An Exploration of Electricity Consumption Patterns is a data analysis project designed to study, analyze, and visualize electricity consumption trends using historical energy usage data. As electricity demand continues to grow due to urbanization, industrialization, and increased dependence on electrical appliances, understanding consumption patterns has become essential for efficient energy management and sustainable development. Traditional methods of analyzing electricity usage often rely on manual reports and disconnected datasets, making it difficult to identify trends, detect anomalies, and make informed decisions.

Energy providers, policymakers, businesses, and consumers frequently face challenges such as unpredictable demand fluctuations, inefficient energy utilization, rising electricity costs, and limited visibility into consumption behavior. These issues can lead to increased operational expenses, energy wastage, and difficulties in planning future electricity requirements.

Users require a centralized and data-driven solution that can analyze electricity consumption patterns, identify usage trends, compare consumption across different time periods or regions, and provide meaningful visualizations for decision-making. The key challenges include scattered data sources, lack of real-time insights, difficulty in identifying peak consumption periods, and limited analytical reporting.

To address these challenges, we propose a comprehensive Electricity Consumption Analysis system that includes:

- Collection and preprocessing of electricity consumption data
- Exploratory Data Analysis (EDA) to identify trends and patterns
- Interactive visualizations of energy usage
- Detection of peak demand periods and consumption anomalies
- Generation of analytical reports to support data-driven decision-making

This data analysis solution simplifies the process of understanding electricity consumption by:

- Centralizing electricity usage data for analysis
- Identifying consumption trends and seasonal variations
- Visualizing energy usage through informative charts and graphs
- Providing actionable insights for efficient energy planning
- Supporting informed decisions through comprehensive analytical reports

By transforming raw electricity consumption data into meaningful insights, the project promotes efficient energy utilization, reduces unnecessary power consumption, and supports sustainable energy management. Its analytical approach enables organizations, utility providers, and researchers to make better decisions for future energy planning and resource optimization.

4.1 Project Overview

Plugging into the Future: An Exploration of Electricity Consumption Patterns is a Tableau-based data visualization and analytics project designed to transform raw electricity consumption data into meaningful and interactive dashboards. By leveraging Tableau's advanced visualization capabilities, the project enables users to analyze electricity usage trends, identify consumption patterns, and gain valuable insights for effective energy management. The dashboard minimizes the complexity of interpreting large datasets by presenting information through intuitive charts, graphs, maps, and key performance indicators (KPIs).

The project incorporates comprehensive **Electricity Consumption Analysis**, where historical electricity usage data is organized and visualized based on various parameters such as time, region, customer category, and consumption levels. Interactive filters and drill-down features allow users to explore data from different perspectives, making it easier to identify seasonal trends, peak demand periods, and variations in electricity usage.

The dashboard also provides **Comparative Analysis**, enabling users to compare electricity consumption across different years, months, regions, or consumer segments. This helps identify areas with high energy demand, evaluate consumption behavior, and support efficient resource planning. Through dynamic visualizations, users can quickly recognize trends and make informed decisions regarding energy distribution and conservation.

To support data-driven decision-making, the project includes interactive dashboards featuring KPIs, trend analysis, consumption forecasts, and performance metrics. These visual reports provide clear insights into electricity usage patterns, helping stakeholders optimize energy utilization, improve operational efficiency, and promote sustainable energy management.

Overall, the Tableau dashboard offers a centralized and user-friendly platform for analyzing electricity consumption data. It empowers utility providers, policymakers, researchers, and energy analysts with actionable insights, enabling better planning, efficient energy allocation, and informed decision-making through powerful data visualization techniques.

IDEATION PHASE

The Ideation Phase forms the foundation of the project by identifying the real-world challenges associated with analyzing electricity consumption data and exploring effective visualization-based solutions. This phase focuses on understanding user requirements, studying available datasets, and designing an interactive Tableau dashboard that converts complex electricity consumption data into meaningful insights. The objective is to develop a solution that supports efficient energy analysis and data-driven decision-making.

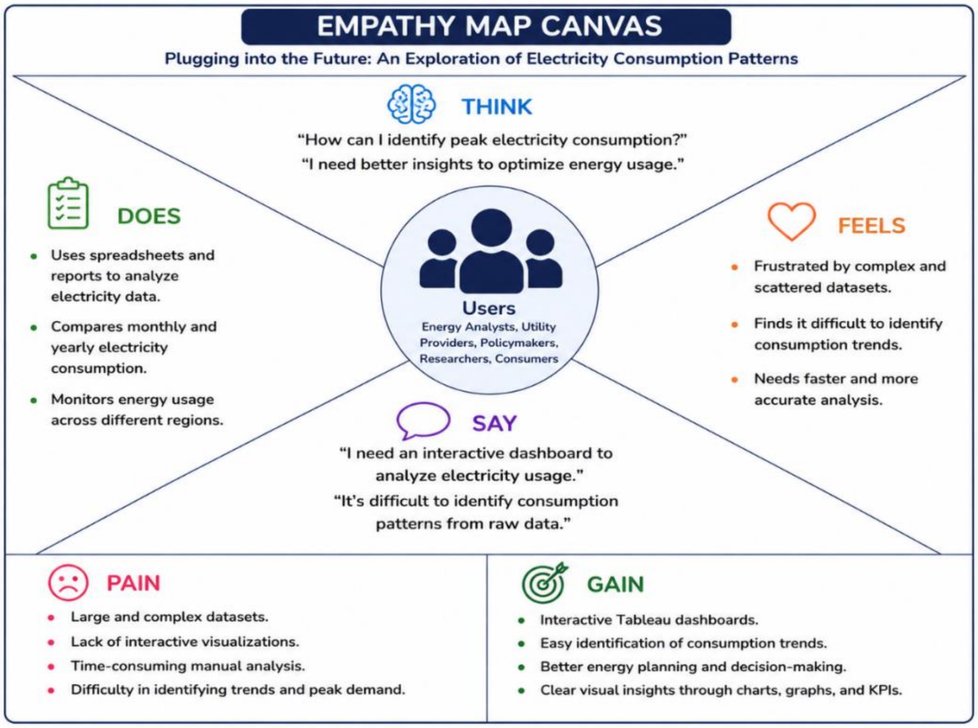
5.1 Problem Statement

Electricity consumption data is generated in large volumes across residential, commercial, and industrial sectors.The absence of interactive visualization tools limits the ability of stakeholders to identify trends, monitor energy usage, and make informed decisions. These challenges include:

- Difficulty in identifying electricity consumption trends over different time periods.
- Limited visibility into regional and sector-wise energy usage.
- Inability to detect peak electricity demand and unusual consumption patterns.
- Time-consuming analysis using static reports and spreadsheets.

5.2 Empathy Map Canvas

To better understand the end-users—mainly inventory managers, medical staff, and administrators—we created an empathy map. This tool helps in viewing the problem from the user's perspective, which is essential for designing user-friendly solutions.



5.3 Brainstorming Solution

Multiple approaches were evaluated before selecting **Tableau** as the primary visualization tool due to its powerful analytics capabilities and interactive dashboard features. The key decisions made during this phase included:

- Use **Tableau Desktop** to create interactive and visually appealing dashboards.
- Connect and preprocess electricity consumption datasets for accurate analysis.
- Design dashboards using charts, graphs, maps, and Key Performance Indicators (KPIs) to represent consumption trends.
- Implement interactive filters, parameters, and drill-down features for detailed exploration of electricity usage.

We also compared Tableau with traditional spreadsheet-based analysis and other visualization tools. Tableau was selected because of its advanced data visualization capabilities, interactive dashboard features

5.4 Goal Setting

The following goals were established to guide the development of the **"Plugging into the Future: An Exploration of Electricity Consumption Patterns"** Tableau Dashboard:

- Develop an interactive and user-friendly **Tableau dashboard** for electricity consumption analysis.
- Visualize electricity consumption trends using charts, graphs, maps, and Key Performance Indicators (KPIs).

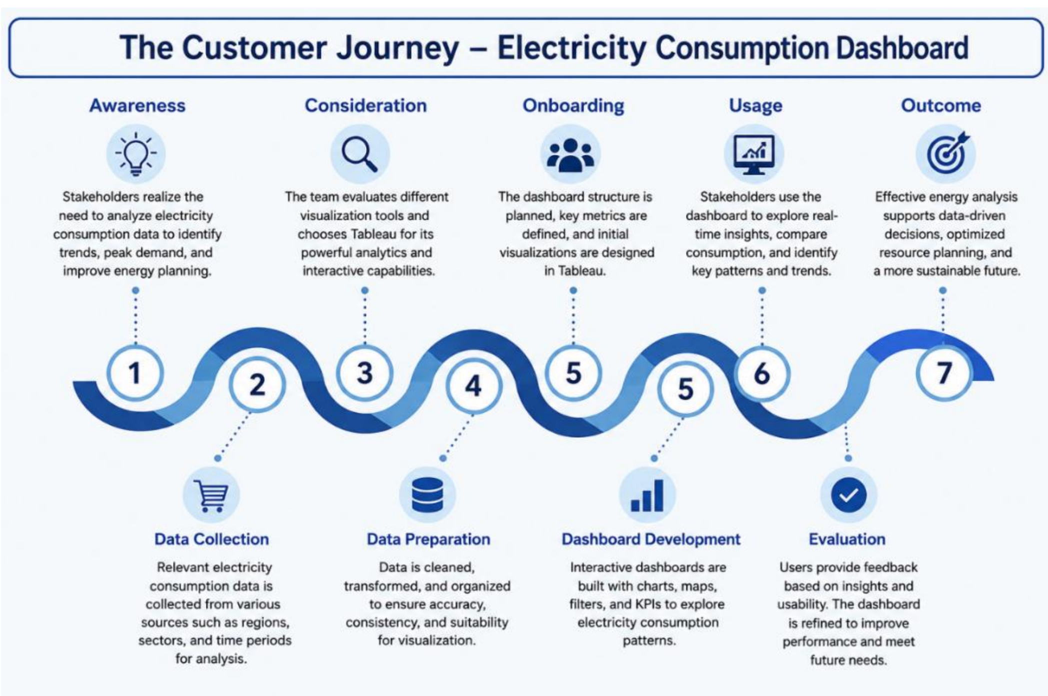
- Enable users to analyze and compare electricity usage across different regions, sectors, and time periods.
- Provide meaningful insights through interactive filters and dashboards to support data-driven decision-making.

5.5 Project Planning

The project was organized into well-defined phases to ensure systematic development and successful completion of the Tableau dashboard. Each phase focused on a specific aspect of the project, beginning with data collection and preparation, followed by data analysis, dashboard design, visualization development, and performance evaluation.

Outcome of the Ideation Phase :

The ideation phase successfully defined the project scope, identified user requirements, and selected **Tableau** as the most suitable platform for data visualization and analysis. It established a clear roadmap for developing an interactive dashboard capable of transforming raw electricity consumption data into meaningful insights. This phase ensured that the project remained user-centric, technically feasible, and aligned with real-world energy analysis and decision-making needs..



REQUIREMENT ANALYSIS

The Requirement Analysis phase focuses on identifying the data, tools, and functionalities required to develop the **"Plugging into the Future: An Exploration of Electricity Consumption Patterns"** Tableau dashboard. This phase ensures that the dashboard effectively analyzes electricity consumption data and presents meaningful insights through interactive visualizations. The requirements were gathered by studying the dataset, understanding user needs, and selecting appropriate visualization techniques.6.1. Functional Requirements

These are the specific functionalities that the system must perform to meet the needs of its users.

6.1 Functional Requirements

The dashboard should provide the following functionalities:

- Import and connect electricity consumption datasets into Tableau.
- Clean and organize the data for accurate analysis and visualization.
- Display electricity consumption trends using interactive charts, graphs, and dashboards.
- Compare electricity usage across different regions, sectors, and time periods.
- Identify peak electricity demand and consumption patterns.
- Provide interactive filters and drill-down options for detailed data exploration.
- Display Key Performance Indicators (KPIs) such as total consumption, average consumption, and highest energy usage.
- Generate visual reports that support data-driven decision-making and energy planning.

6.2 Non-Functional Requirements

The dashboard should satisfy the following quality requirements:

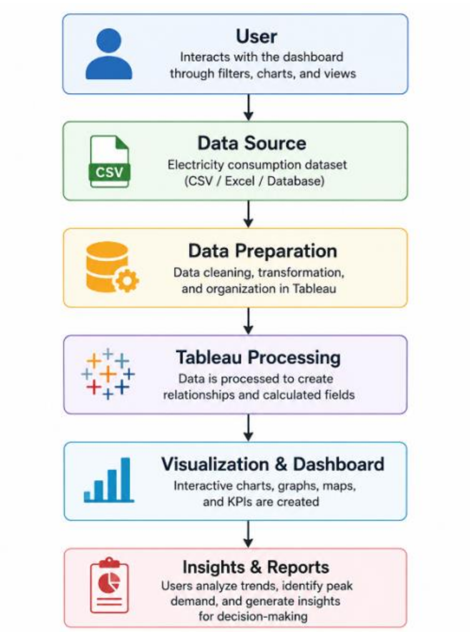
- **Usability:** Provide an intuitive and user-friendly interface that is easy to navigate.
- **Performance:** Load and display visualizations efficiently, even for large datasets.
- **Reliability:** Ensure accurate representation of electricity consumption data.
- **Scalability:** Support future datasets and additional visualizations without major redesign.
- **Maintainability:** Allow easy updates to datasets and dashboard components.
- **Compatibility:** Operate seamlessly on Tableau Desktop and Tableau Public/Server platforms.

6.3 Data Flow Diagram (Level 0)

Raw electricity consumption data is imported from datasets into Tableau, where it undergoes preprocessing and visualization.

Data Flow:

- Electricity consumption dataset is imported into Tableau.
- Data is cleaned, organized, and prepared for visualization.
- Tableau processes the data and creates interactive charts, graphs, maps, and KPIs.



6.4 Technology Stack

The project is developed using **Tableau** as the primary data visualization tool, supported by data preprocessing and analysis techniques. The technology stack enables efficient data handling, interactive dashboard creation, and meaningful visualization of electricity consumption patterns:

Component	Technology	Purpose
Visualization Platform	Tableau Desktop	Develop interactive dashboards and visualizations
Data Source	CSV / Excel Dataset	Store electricity consumption data for analysis
Data Preparation	Tableau Data Source & Data Cleaning	Clean, organize, and transform data before visualization
Visualization	Charts, Graphs, Maps, KPIs	Display electricity consumption trends and patterns

Component	Technology	Purpose
Interactivity	Filters, Parameters, Actions	Enable dynamic exploration and comparison of data
Analytics	Calculated Fields & Table Calculations	Perform analytical computations and derive insights
Dashboard Publishing	Tableau Public / Tableau Server	Publish and share interactive dashboards with users

6.4.1. Custom Objects & Fields

- **Type:** Data Source (CSV / Excel Dataset)
- **Purpose:** The dataset serves as the primary source of information for analyzing electricity consumption patterns. It contains the attributes required to create interactive dashboards and visualizations in Tableau.
- **Key Data Fields:**
    - **Date:** Records the time period of electricity consumption.
    - **Region:** Represents the geographical area where electricity is consumed.
    - **Consumer Category:** Identifies the type of consumer (Residential, Commercial, or Industrial).
    - **Electricity Consumption (kWh):** Indicates the amount of electricity consumed.
    - **Peak Demand:** Represents the highest electricity demand during a specific period.
    - **Energy Cost:** Displays the cost associated with electricity consumption.

- **Why Used:** These data fields enable Tableau to perform efficient analysis, generate interactive visualizations, compare electricity usage across different dimensions, and provide meaningful insights for energy management and decision-making.

PROJECT DESIGN

The **"Plugging into the Future: An Exploration of Electricity Consumption Patterns"** project is designed to analyze electricity consumption data using interactive Tableau dashboards. The project transforms raw electricity consumption data into meaningful visualizations, enabling users to identify trends, compare regional consumption, and make informed decisions regarding energy management.

Traditional electricity consumption reports are often presented in spreadsheets or static documents, making analysis difficult and time-consuming. This Tableau dashboard addresses these challenges by providing dynamic visualizations, interactive filters, and comparative analysis across different years, regions, states, and time periods.

The project consists of multiple dashboards that collectively provide a comprehensive view of electricity consumption across India.

7.1 The Need

Growing electricity demand requires efficient monitoring and analysis to support sustainable energy planning. Traditional reporting methods fail to provide interactive insights and make it difficult to identify important consumption trends.

The proposed Tableau dashboard addresses these challenges by:

1. **Analyzing Electricity Consumption Trends**
    - Visualize monthly, quarterly, yearly, and regional electricity consumption.
2. **Supporting Data-Driven Decisions**
    - Help policymakers and utility providers identify high and low consumption areas.
3. **Interactive Data Exploration**
    - Allow users to filter data by year, state, region, and month.
4. **Comparative Analysis**
    - Compare electricity usage across different years and geographical regions.
5. **Improved Visualization**
    - Convert complex datasets into easy-to-understand charts, graphs, maps, and dashboards.

7.2 Proposed Solution

To overcome the limitations of traditional reporting systems, a **Tableau-based Electricity Consumption Analysis Dashboard** has been developed.

Key Features

- Interactive Tableau Dashboards
- Monthly and Quarterly Consumption Analysis
- State-wise Electricity Consumption Maps
- Region-wise Consumption Comparison
- Top N and Bottom N State Analysis
- Year-wise Consumption Trends
- Dynamic Filters for Year, Region, Month, and State
- Interactive Charts, Maps, Pie Charts, Line Charts, and Bar Charts

Benefits

- Simplifies electricity consumption analysis.
- Helps identify peak and low consumption regions.
- Supports effective energy planning.
- Enables quick comparison between different years.
- Provides interactive dashboards for better decision-making.

7.3 Solution Architecture

The project follows a layered architecture for efficient data processing and visualization.

1. Data Layer

- Electricity Consumption Dataset (CSV/Excel)
- Data containing Year, Month, Quarter, Region, State, and Usage values.

2. Data Preparation Layer

- Import data into Tableau.
- Clean and transform the dataset.
- Create calculated fields and relationships.

3. Visualization Layer

Interactive dashboards created using:

- Bar Charts
- Line Charts
- Pie Charts
- Geographic Maps
- KPI Indicators
- Top N and Bottom N Analysis

4. User Interaction Layer

Users interact with the dashboard using:

- Year Filter
- Region Filter
- State Filter
- Month Filter
- Dashboard Navigation Buttons

5. Reporting Layer

The final dashboards provide:

- Monthly Consumption Reports
- Quarterly Consumption Reports
- Region-wise Analysis
- State-wise Analysis
- Year-wise Comparison
- Top N and Bottom N Consumption Reports

Dashboard Overview

Dashboard 1 – Consumption Trend Analysis

- Quarterly Electricity Consumption
- Monthly Consumption (2019)
- Monthly Consumption (2020)
- Year-wise Usage Trend

Purpose:
To analyze electricity consumption trends over different months, quarters, and years.

Dashboard 2 – Geographic Consumption Analysis

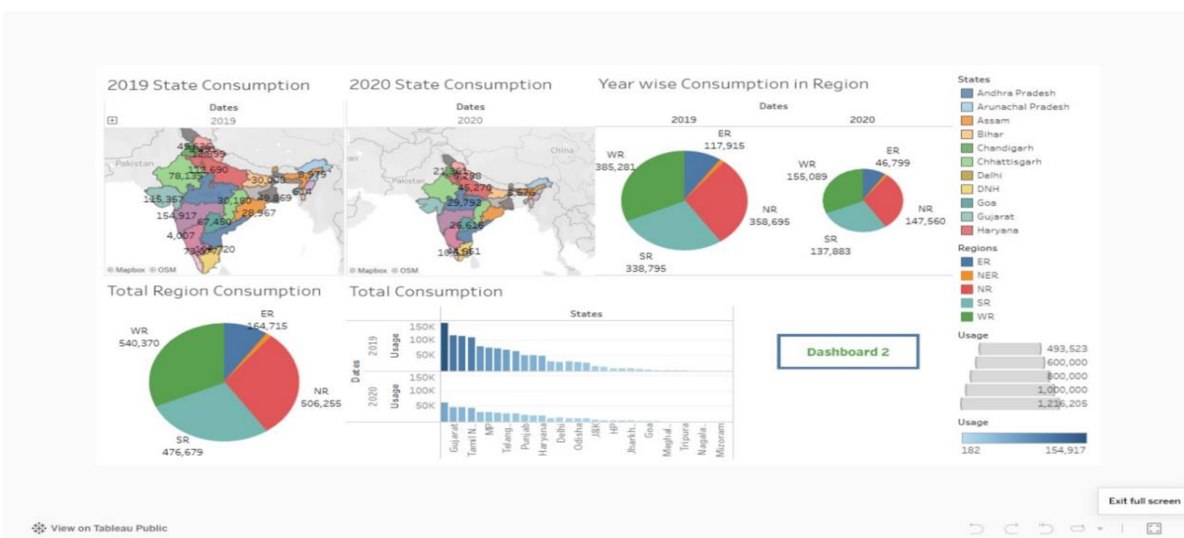
- India State-wise Consumption Map (2019)
- India State-wise Consumption Map (2020)

- Region-wise Consumption Pie Charts
- Total State Consumption Comparison

Purpose:
To visualize electricity consumption geographically and compare regional energy usage.

Dashboard 3 – Regional Performance Analysis

- Region-wise State Consumption
- Top N Highest Electricity Consuming States
- Bottom N Lowest Electricity Consuming States
- Dynamic Top N and Bottom N Parameters



Purpose:
To identify the highest and lowest electricity-consuming states for comparative analysis

PROJECT PLANNING & SCHEDULING

Effective planning and scheduling were essential for the successful development of the **"Plugging into the Future: An Exploration of Electricity Consumption Patterns"** Tableau Dashboard. The project was executed in a structured manner, ensuring that each phase—from data collection and preparation to dashboard development, testing, and documentation—was completed within the planned timeline.

8.1 Project Planning Objectives

- Define milestones for each stage of dashboard development.
- Allocate sufficient time for data collection, cleaning, visualization, and testing.
- Ensure a systematic workflow from planning to deployment.
- Deliver an interactive and user-friendly Tableau dashboard within the scheduled timeline.

8.2 Project Development Phases & Milestones

Milestone No.	Milestone Name	Activities Covered	Timeline
M1	Dataset Collection	Collect electricity consumption dataset from reliable sources	Day 1
M2	Data Cleaning	Remove missing values, correct inconsistencies, prepare dataset	Days 2–3
M3	Data Preparation	Create calculated fields, organize dimensions and measures	Day 4
M4	Dashboard Planning	Identify KPIs and select suitable visualizations	Days 5–6
M5	Dashboard Development	Create charts, maps, graphs, and worksheets in Tableau	Days 7–10
M6	Interactive Features	Add filters, parameters, dashboard actions, and navigation	Days 11–12
M7	Dashboard Integration	Combine worksheets into interactive dashboards	Days 13–14
M8	Dashboard Testing	Verify calculations, filters, and visualizations	Days 15–16
M9	Performance Optimization	Improve dashboard responsiveness and layout	Day 17
M10	Documentation	Prepare project report and capture dashboard screenshots	Days 18–19
M11	Final Review	Review dashboard functionality and documentation	Day 20

8.3 Tools Used for Planning and Execution

Tool	Purpose
Tableau Desktop	Dashboard development and visualization
Microsoft Excel / CSV	Data storage and preprocessing
Tableau Public	Dashboard publishing and sharing
Draw.io / Lucidchart	Architecture and flow diagrams
Microsoft Word	Documentation and report preparation
Microsoft PowerPoint	Project presentation
GitHub (Optional)	Version control and project backup

8.4 Key Success Indicators

- Interactive Tableau dashboards developed successfully.
- Accurate visualization of electricity consumption data.
- Proper functioning of filters, parameters, and dashboard actions.
- Effective comparison of electricity usage across years, regions, and states.
- Meaningful insights generated through charts, maps, and KPIs.
- Project completed within the planned timeline.
- Documentation and dashboard validated successfully.

FUNCTIONAL AND PERFORMANCE TESTING

9.1 Functional Testing

Functional testing verifies that every component of the **"Plugging into the Future: An Exploration of Electricity Consumption Patterns"** Tableau dashboard performs according to the specified requirements. The testing process ensures that all visualizations, filters, parameters, and dashboard interactions function correctly and provide accurate analytical insights.

Test Objectives

- Verify that all dashboards load successfully without errors.
- Ensure charts, graphs, maps, and KPIs display accurate electricity consumption data.
- Validate the functionality of filters and parameters.
- Confirm that navigation between dashboards works correctly.
- Verify that calculated fields and aggregations produce accurate results.
- Ensure visualizations respond correctly to user interactions.

Functional Test Cases

Test Case	Description	Expected Result	Status
TC01	Open Dashboard 1	Dashboard loads successfully	Pass
TC02	Apply Year Filter (2019/2020)	Charts update according to selected year	Pass
TC03	Apply Region Filter	Displays data for the selected region	Pass
TC04	Verify Monthly & Quarterly Charts	Charts display correct electricity consumption values	Pass
TC05	Verify State-wise Maps	Maps correctly visualize state consumption	Pass
TC06	Test Top N Parameter	Displays the selected number of highest consuming states	Pass
TC07	Test Bottom N Parameter	Displays the selected number of lowest consuming states	Pass
TC08	Dashboard Navigation	Navigation buttons switch between dashboards correctly	Pass
TC09	Verify Calculated Values	KPIs and aggregated values match the dataset	Pass
TC10	Verify Tooltips	Tooltip displays accurate information on hover	Pass

Dashboard Validation Testing

Component	Tested For	Result
Dashboard 1	Monthly, Quarterly and Year-wise Analysis	Pass
Dashboard 2	State-wise Maps and Region-wise Consumption	Pass
Dashboard 3	Top N, Bottom N and Region-wise Analysis	Pass

User Interface Testing

- Dashboard layout is clear and easy to understand.
- Charts and maps are displayed correctly without overlap.
- Filters and parameters respond accurately to user selections.
- Navigation buttons switch between dashboards successfully.

- Legends, labels, and tooltips improve data readability.

9.2 Performance Testing

Performance testing evaluates the responsiveness and efficiency of the Tableau dashboard while interacting with electricity consumption data.

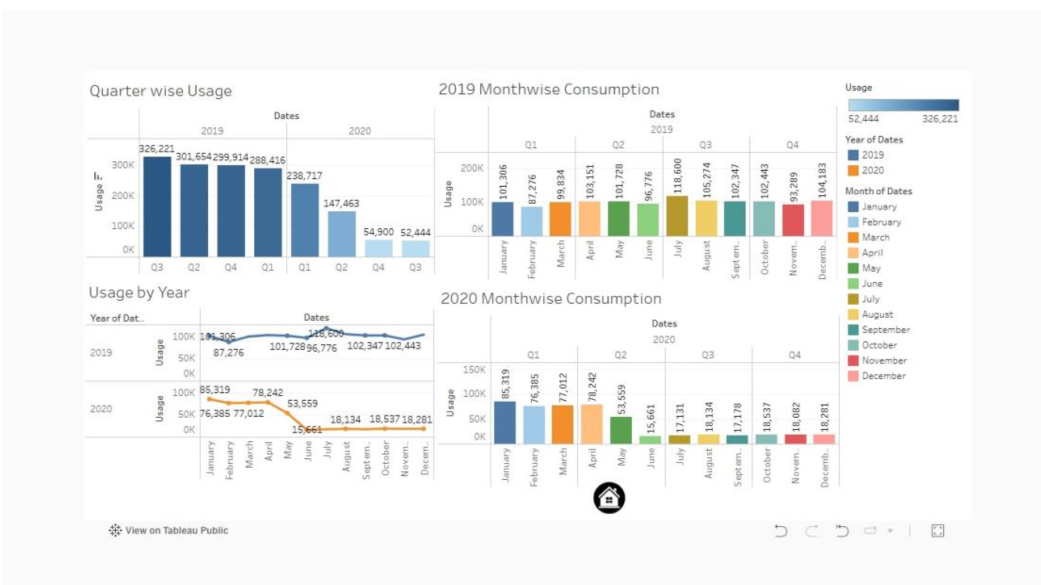
Test Case	Description	Expected Result	Status
PT01	Dashboard Loading	Dashboard loads within a few seconds	Pass
PT02	Filter Response	Charts update immediately after filter selection	Pass
PT03	Parameter Response	Top N and Bottom N parameters update correctly	Pass
PT04	Map Rendering	Geographic maps load without delay	Pass
PT05	Chart Rendering	All charts display correctly without distortion	Pass
PT06	Dashboard Navigation	Navigation between dashboards is smooth	Pass

Conclusion

Functional and performance testing confirms that the Tableau dashboard operates as intended. All dashboards, filters, parameters, maps, and visualizations function correctly, providing accurate electricity consumption analysis. The dashboard responds efficiently to user interactions, enabling meaningful exploration of electricity consumption patterns and supporting effective, data-driven decision-making. The project is therefore ready for presentation and deployment.

10.RESULTS

10.1 Dashboard 1 – Time-wise Electricity Consumption Analysis

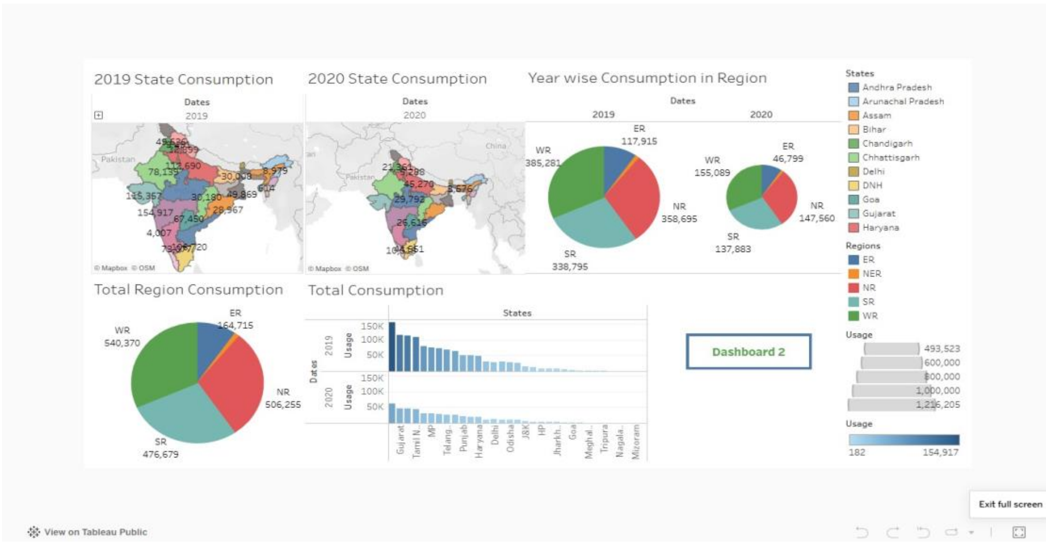


Result:

- Displays monthly, quarterly, and yearly electricity consumption.
- Compares electricity usage between **2019** and **2020**.
- Identifies peak and low consumption months.

- Helps users understand seasonal electricity usage trends.

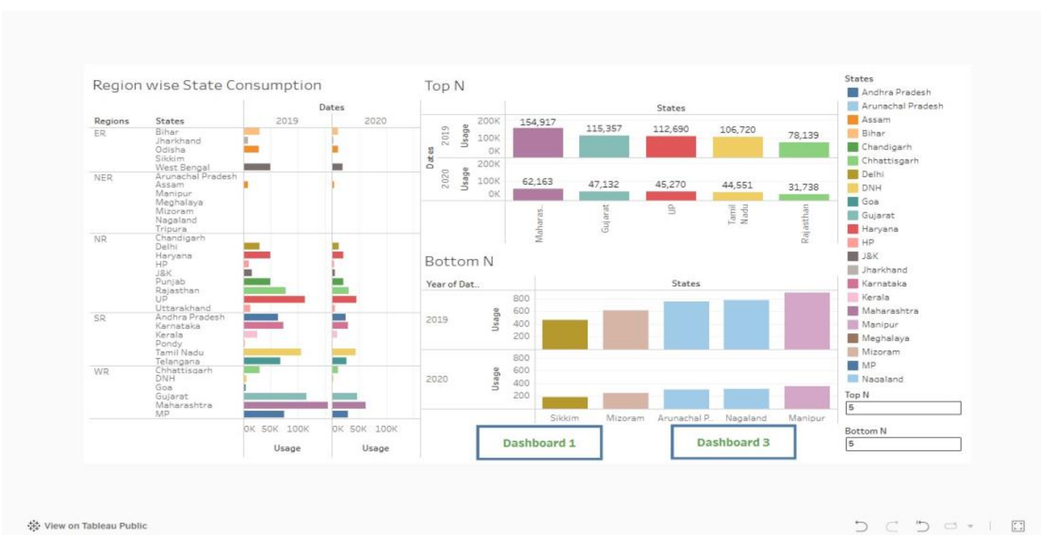
10.2 Dashboard 2 – State-wise and Region-wise Analysis



Result:

- Visualizes electricity consumption across Indian states.
- Compares regional electricity usage.
- Highlights regions with high and low electricity demand.
- Supports geographical analysis using interactive maps and pie charts.

10.3 Dashboard 3 – Top N & Bottom N State Analysis



Result:

- Displays the highest electricity-consuming states.
- Displays the lowest electricity-consuming states.
- Provides region-wise comparison.

- Allows users to dynamically change **Top N** and **Bottom N** values using parameters.

ADVANTAGES & DISADVANTAGES

Advantages

1. Interactive Data Visualization

Presents electricity consumption data through interactive charts, maps, and dashboards, making complex information easier to understand.

2. Better Decision-Making

Provides meaningful insights into electricity consumption trends, helping stakeholders make informed energy management decisions.

3. Real-Time Data Analysis

Allows users to instantly analyze and compare electricity consumption using filters, parameters, and interactive visualizations.

4. Easy Trend Identification

Helps identify monthly, quarterly, yearly, and regional electricity consumption patterns quickly and accurately.

5. User-Friendly Dashboard

Offers an intuitive interface with interactive filters, maps, and charts, enabling users to explore data without technical expertise.

6. Comprehensive Reporting

Generates visual reports that summarize electricity consumption across different regions, states, and time periods.

7. Efficient Data Exploration

Supports dynamic filtering and drill-down analysis, allowing users to focus on specific years, regions, or states.

8. Supports Energy Planning

Provides valuable insights that assist utility providers, researchers, and policymakers in planning efficient energy distribution and conservation strategies.

Disadvantages

1. Data Quality Dependency

The accuracy of dashboard insights depends on the quality and completeness of the input dataset.

2. Limited Real-Time Updates

The dashboard reflects the available dataset and requires periodic data refreshes unless connected to a live data source.

3. Tableau Software Requirement

Creating and modifying dashboards requires Tableau Desktop or access to Tableau Server/Public.

4. Learning Curve

New users may require basic training to understand Tableau features and interactive dashboard functionalities.

5. Large Dataset Performance

Very large datasets may increase dashboard loading time and reduce performance if not optimized.

Summary

Aspect	Advantages	Disadvantages
Visualization	Interactive charts, maps, and dashboards	Requires familiarity with Tableau
Data Analysis	Quick identification of trends and patterns	Depends on accurate and updated data
Decision Support	Helps in effective energy planning	Results are limited by available data
User Experience	Easy-to-use interface with interactive filters	Some users may require initial training
Performance	Fast analysis for optimized datasets	Large datasets may affect dashboard speed
Accessibility	Dashboards can be shared through Tableau Public/Server	Editing requires Tableau Desktop

CONCLUSION

The **"Plugging into the Future: An Exploration of Electricity Consumption Patterns"** project successfully demonstrates how Tableau can transform raw electricity consumption data into meaningful and interactive visualizations. By integrating multiple dashboards with charts, maps, filters, and performance indicators, the project enables users to analyze electricity consumption trends across different years, regions, and states efficiently.

The dashboard addresses the challenges of traditional data analysis by providing:

- Interactive visualization of electricity consumption patterns.
- Comparative analysis across months, quarters, years, regions, and states.
- Dynamic filtering for detailed exploration of data.
- Geographic visualization using state-wise maps.
- Meaningful insights through charts, graphs, and Key Performance Indicators (KPIs).

The project highlights the effectiveness of Tableau as a business intelligence and data visualization tool for analyzing large datasets. The interactive dashboards simplify complex electricity consumption data, enabling users to identify consumption trends, compare regional performance, and support informed decision-making for efficient energy management.

In conclusion, this Tableau-based dashboard provides an efficient, user-friendly, and visually appealing solution for electricity consumption analysis. It supports utility providers, policymakers, researchers, and energy analysts in understanding consumption patterns, improving energy planning, and promoting sustainable resource utilization through data-driven insights.

FUTURE SCOPE

The current Tableau dashboard provides an effective platform for analyzing and visualizing electricity consumption patterns across different regions and time periods. However, several enhancements can be incorporated in the future to improve its analytical capabilities, user experience, and decision-making support.

1. Real-Time Data Integration

- Integrate the dashboard with live electricity consumption databases or APIs.
- Enable automatic data refresh for real-time monitoring and analysis.

2. Predictive Analytics

- Incorporate machine learning models to forecast future electricity consumption based on historical trends.
- Help utility providers and policymakers anticipate future energy demand.

3. Advanced Interactive Dashboards

- Introduce additional drill-down features, dynamic parameters, and custom user controls.
- Enable more detailed exploration of electricity consumption at district or city levels.

4. Mobile Dashboard Accessibility

- Optimize the dashboard for smartphones and tablets using Tableau Mobile.
- Allow users to access electricity consumption insights anytime and anywhere.

5. Renewable Energy Analysis

- Extend the dashboard to include renewable energy sources such as solar, wind, and hydroelectric power.
- Compare renewable and conventional energy consumption patterns.

6. Enhanced Geographic Visualization

- Develop more detailed GIS-based maps for district-wise and city-wise electricity consumption analysis.
- Improve regional planning through advanced spatial visualization techniques.

7. Energy Efficiency Recommendations

- Incorporate analytical models that provide suggestions for reducing electricity consumption.
- Support sustainable energy utilization by identifying high-consumption areas and recommending optimization strategies.

GitHub & Project Demo Link : https://github.com/Srividya2315/Medical-Inventory-Management.git